

for rank-based trimming. In practice, the choice of α can be based on the proportion of outliers observed in a boxplot or histogram of the D_i^2 s.

The estimator $\hat{\Theta}$ defined above belongs to the general class of M-estimators, which have well-known asymptotic properties (Van der Vaart, 1998, ch. 5). As shown in the Appendix, $\sqrt{n}\{\text{vec}(\hat{\Theta}) - \text{vec}(\Theta_0)\}$ follows an approximate $N(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1})$ distribution for large n , with

$$\mathbf{A} = 2\mathbb{E}\{\rho''(e)w^2(\mathbf{U})\Sigma_0^{-1}\mathbf{R}\mathbf{R}^T \otimes \mathbf{U}\mathbf{U}^T\} + \mathbf{I}_q \otimes \mathbb{E}\{\rho'(e)w(\mathbf{U})\mathbf{U}\mathbf{U}^T\}, \quad (15)$$

$$\mathbf{B} = \mathbb{E}\left[\{\rho'(e)\}^2 w^2(\mathbf{U})\mathbf{R}\mathbf{R}^T \otimes \mathbf{U}\mathbf{U}^T\right]. \quad (16)$$

The matrices $\mathbf{A}$ and $\mathbf{B}$ can be easily estimated, replacing expectations by averages. This asymptotic distribution can be used, for instance, to test significance of the regression: if $\Theta_0 = \mathbf{0}$, Wald's statistic $Q = n\text{vec}(\hat{\Theta})^T \hat{\mathbf{A}}\hat{\mathbf{B}}^{-1}\hat{\mathbf{A}}\text{vec}(\hat{\Theta})$ follows an approximate χ_{pq}^2 distribution for large n , so we decide the regression is significant if $Q \geq \chi_{pq,1-\alpha}^2$ for a given level α . We can also construct marginal tests and confidence intervals for the individual coefficients θ_{kl} .

In Section 3 we will study the accuracy of this asymptotic approximation. It is our experience that the distribution of $\hat{\Theta}$ approaches normality quite fast, but the above “sandwich formula” tends to underestimate the variance when the sample size n is small. In that case it is better to use bootstrap estimators of the covariance matrix of $\text{vec}(\hat{\Theta})$.

3 Simulations

In this section we study by simulation the finite-sample behavior of the estimators (10). To this end, we generated data from model (8) with $\mathbf{U} \sim N(\mathbf{0}, \mathbf{\Lambda})$ and $\mathbf{W} \sim N(\mathbf{0}, \mathbf{\Sigma})$, where